

mini assignment momentum II

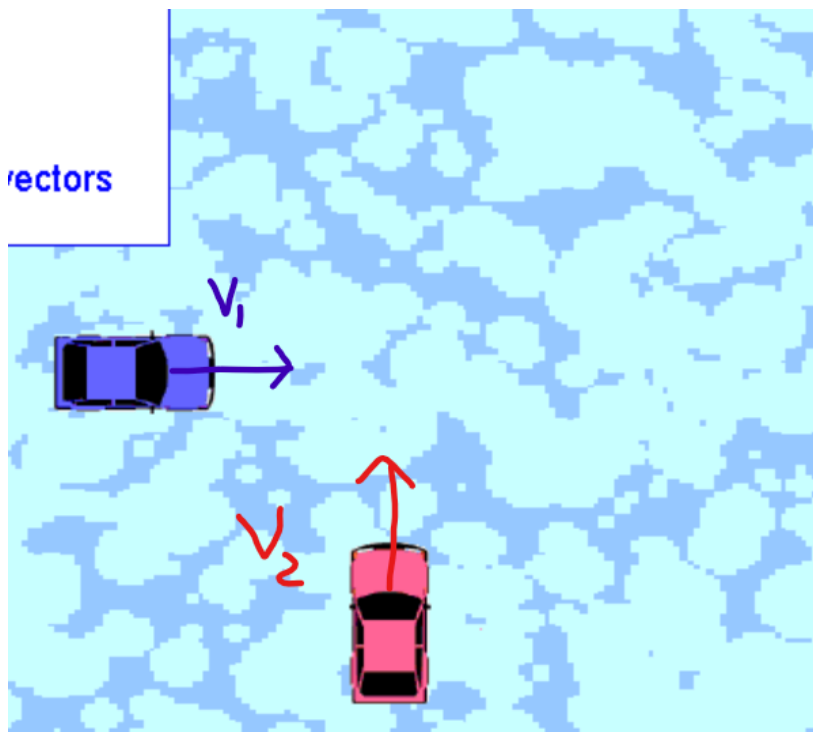
 This is a preview of the published version of the quiz

Started: Apr 29 at 8:02pm

Quiz Instructions



Question 1 12.5 pts



The 2 cars collide with each other and stick together. In which direction do they move after collision?
There is no friction. momentum is conserved.

hint: momentum is a vector

☐

Along the same direction as the red before collision

☐

at 45 degrees above the direction of the blue star before the collision

☐

they stop on the spot

☐

Along the same direction as the blue before collision

☐

Question 2 12.5 pts

A lamborghini Urus speeds up from 30mph to 90mhp. Its kinetic energy will increase by a factor of

☐

9

☐

6

☐

2

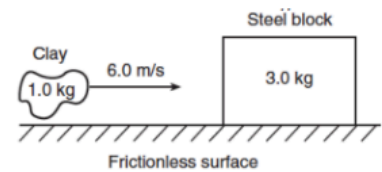
☐

3

☐

Question 3 12.5 pts

3. A 3.0 kg steel block is at rest on a frictionless horizontal surface. A 1.0 kg lump of clay propelled horizontally at 6.0 m/s toward the block as shown in the diagram. Upon collision, the clay and steel block stick together and move to the right with a speed of:



(A) 1.5 m/s

(B) 2.0 m/s

(C) 3.0 m/s

(D) 6.0 m/s

☐

A

☐

D

☐

B

☐

C

☐

Question 4 12.5 pts

8. Two spacemen are floating together with zero speed in a gravity-free region of space. The mass of spaceman A is 120 kg and that of spaceman B is 90 kg. Spaceman A pushes B away from him with B attaining a final speed of 0.5m/s. The final recoil speed of A is:

- ☐ A zero
- ☐ B 0.38 m/s
- ☐ C 0.5 m/s
- ☐ D 0.67 m/s
- ☐ E 1.0 m/s

(speed is a scalar. no direction. So just the magnitude)

☐ A

☐ D

☐ C

☐ B

⋮

Question 5 12.5 pts

Compute the change in momentum for A and then for B (mass x change in velocity).

Compare the magnitude of the change in momentum for each spaceman.

☐

they are equal to each other

☐

B has a larger speed, it has a larger change in momentum

☐

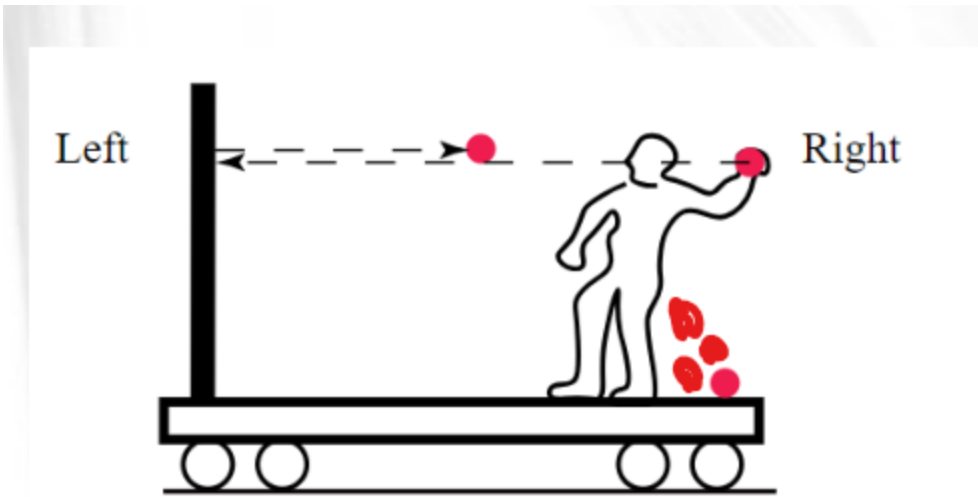
I spaced out in class

☐

A has more mass, it has a larger change in momentum

⋮

Question 6 12.5 pts



you have a stack of bouncing balls (red) that you throw. They bounce off the wall and you don't catch them.

What happens? (conservation momentum. velocity is a vector)

- ☐ the platform is moving at left
- ☐ the platform is not moving.
- ☐ The platform is moving at right
- ☐ the platform jumps up



Question 7 12.5 pts

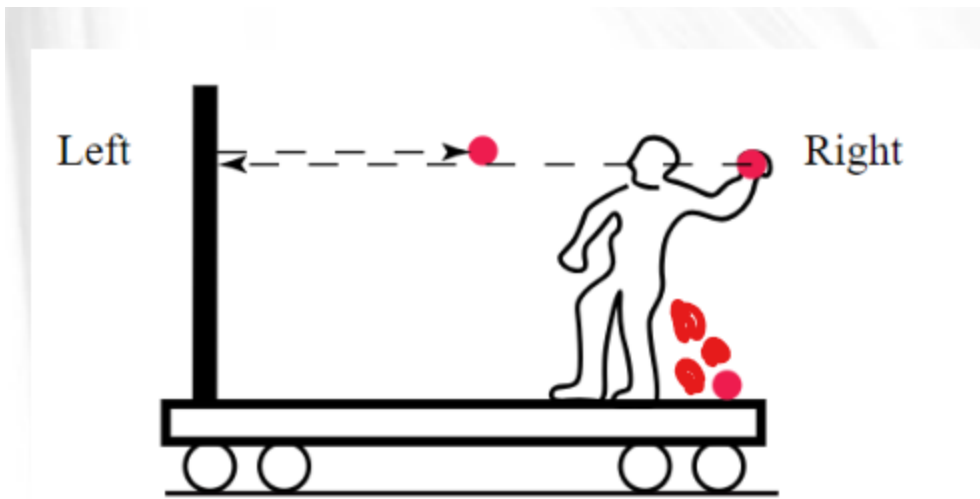
Building on the previous question.

Now you catch the ball after it bounces off the wall. So you are using only one ball.

- ☐ the platform is moving at right
- ☐ the platform is moving at left
- ☐ the platform is not going anywhere
- ☐ I missed the class and didn't catch up



Question 8 12.5 pts



You have a stack of bouncing ball and thrown them against the wall. one after another. you dont catch them. The thrust of the platform depends on

(the net force acting on the platform)

- ☐ only on of the speed of the throw
- ☐ the color of the ball. blue works better
- ☐ only on the amount of balls you have stacked
- ☐ the speed of the throw and the number of balls thrown per unit time

Not saved

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